EXPERIMENT 13

Case Study on Low-Code Limitations

AIM

To analyze the architectural and operational limitations that may arise when developing and maintaining the Patient Appointment Management System using Zoho Creator.

APPLICATION CONTEXT

The Patient Appointment Management System uses forms, reports, workflows, analytics, and integrations. Low-code development can speed up delivery, but platform constraints should be considered as requirements grow.

1. Overview

Zoho Creator makes it possible to build the appointment form, reports, dashboard, and workflows with less manual coding. The same platform-based approach can create constraints when requirements become highly customized or require extensive integration.

2. Key Limitations

- Platform dependency and portability: forms, workflows, and Deluge logic are tied to the platform, so moving the complete application elsewhere may require redesign and rewriting.
- Workflow and execution limits: complex or repeated processing can be constrained by platform execution limits. Use targeted criteria and move computationally intensive tasks to a suitable external service when necessary.
- Data and query control: the platform abstracts much of the underlying data layer. Developers have less direct control over indexing and low-level query tuning than in a fully custom database application.
- User-interface customization: visual builders are convenient for standard forms and reports, but highly specialized layouts or interactions may require custom components.
- Plan and service availability: some advanced security, deployment, analytics, or integration features may depend on account configuration or subscription.

3. Impact on the Appointment System

These limitations could affect appointment scheduling rules, large appointment histories, complex analytics, custom patient portals, and integrations with external hospital or notification services.

4. Mitigation Strategies

- Use specific report filters and criteria-based lookups instead of unnecessary broad queries.
- Keep Deluge workflows small, readable, and conditional where possible.
- Use documented APIs for external integrations and test authentication and error handling.
- Apply role-based permissions and review access to patient records.
- Maintain a clear export, backup, and migration plan, and verify which features are available in the account before depending on them.

5. Conclusion

Low-code development is effective for rapidly building a practical appointment-management solution. A careful design, targeted workflows, suitable permissions, and a plan for integrations and migration help reduce the impact of platform limitations.

RESULT

Thus, the case study summarizes important limitations of low-code development and suggests practical ways to manage them.